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Regular Article

From fear to adaptation: The dynamic impact of AI on worker behavior and technological well-being

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ABSTRACT

The rapid rise of generative AI, marked by ChatGPT's launch, has reshaped the labor market by intensifying technological substitution and prompting workers to adjust their behavior and mindset. This study applies a Difference-in-Differences model to examine how AI substitution concerns have evolved in Taiwan's labor market. Using national survey data from 2020 (before ChatGPT), 2022 (its launch year), and 2023 (one year after), it explores differences across skill levels and industries, offering a broad view of AI's labor market impact. The findings show that workers most concerned about AI replacement initially increased their use of digital skills during ChatGPT's adoption phase, but this adjustment weakened over time. Their technological well-being followed a pattern of short-term decline, mid-term recovery, and long-term divergence. Service sector workers, more reliant on digital tools, adjusted faster and adapted more effectively than manufacturing workers, who exhibited slower, steadier changes. Skill-level analysis reveals that high-skill workers quickly adapted to technological demands and saw improvements in well-being. In contrast, low-skill workers struggled significantly, highlighting the need for urgent skill retraining. This study fills a critical gap by adopting a dynamic approach to examine skill differentiation, industry disparities, and psychological adaptation to AI. It recommends that policymakers focus on nationwide skill retraining and psychological support programs to help workers adapt. These measures are essential for balancing technological progress, labor market stability, and workforce development.

1. Introduction

The rapid rise of Artificial Intelligence (AI) is transforming labor markets worldwide. AI improves productivity, optimizes resource use, and drives innovation. However, its effects vary across industries and skill levels, creating opportunities and challenges (Furman & Seamans, 2019; Song, 2024). While AI-driven substitution pressures low-skilled workers, high-skilled workers benefit from its complementary effects (Damioli et al., 2023). This growing labor market divide has heightened academic interest in AI's impact on employment and skill adaptation.

The development of generative AI, especially with ChatGPT, represents a significant advance in natural language processing and human-machine interaction. Chen et al. (2023) highlight that generative AI creates new job roles and reshapes skill requirements for existing ones. While its potential is widely acknowledged, research shows a dual impact: it drives growth in high-skilled, tech-focused jobs while increasing substitution pressures on low-skilled, routine-based work (Owolabi et al., 2024; Song, 2024). This shift initially raises worker anxiety and unemployment risks but ultimately leads to significant labor market adjustments.

Taiwan's labor market features a distinctive structural profile that provides important context for this study. Over the past decades, Taiwan has become heavily dependent on high-technology manufacturing exports (especially semiconductors and electronics), making it one of the world's leading high-tech export economies. The service sector now employs a majority of the workforce (roughly 60 %), reflecting a shift toward services in employment. Educational attainment is relatively high; many workers have completed tertiary or vocational education, yet skills are unevenly distributed, contributing to occupational polarization. Like many advanced economies, Taiwan has experienced substantial growth in high-skill, knowledge-intensive jobs alongside a hollowing-out of routine mid-skill occupations, which widens the gap between high-skill and low-skill employment opportunities.

Despite growing research on AI's impact on the labor market, key gaps remain. Most studies examine static effects, overlooking the dynamic changes driven by generative AI advancements. Existing research

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focuses mainly on developed economies, offering limited insight into their varied effects across emerging economies and industries. There is little empirical evidence on how workers psychologically adapt and respond to AI substitution pressures, especially regarding subjective technological well-being. As generative AI, such as ChatGPT, continues to expand, deeper analysis is needed to understand its specific labor market effects and the adaptation processes it triggers.

Additionally, classical technology adoption frameworks (e.g., the Technology Acceptance Model) have largely emphasized rational factors (perceived usefulness and ease of use) in technology uptake, offering limited insight into the emotional and cognitive turbulence (e.g., heightened risk aversion) that can accompany disruptive innovations. The initial fear, psychological stress, and uncertainty sparked by AI may fundamentally alter workers' adaptation pathways—elements not fully addressed by existing models. By examining how perceived control and self-efficacy dynamically evolve in the face of AI disruption, this study proposes an enriched conceptual perspective to capture the full spectrum of worker responses from initial fear to eventual adaptation and divergence.

This study examines Taiwan's labor market to address these gaps, focusing on workers' perceptions of job displacement and behavioral adaptations before and after AI adoption, particularly ChatGPT. Taiwan offers a meaningful context for broader insights due to its high digital penetration, advanced manufacturing and service sectors, active government AI initiatives, and diverse workforce composition. Using data from three points—2020 (pre-ChatGPT), 2022 (ChatGPT's launch year), and 2023 (post-launch)—, it analyzes changes in AI-induced job substitution concerns and their impact on digital tool usage across skill levels and industries. Additionally, the study incorporates subjective technological well-being to assess the psychological effects of AI disruptions. Comparing responses among high-, medium-, and low-skilled workers and between the manufacturing and service sectors provides a comprehensive analysis of AI's varied labor market effects.

The empirical results show that concerns about AI-driven job displacement significantly shape workers' behavioral adaptations, following distinct phases. Before and during ChatGPT's launch, highly concerned workers increased their use of computers and the internet. However, this effect became statistically insignificant a year later, possibly due to job disruptions or reduced motivation to adapt. In the short term, these workers experienced a decline in subjective technological well-being. Over time, adaptive behaviors and psychological adjustments led to a gradual recovery, highlighting a dynamic adaptation process to AI disruptions.

Industry-specific results show that service sector workers adapted more quickly to ChatGPT's introduction. Those highly concerned about AI-driven displacement increased their computer and internet usage by 10.496 % during the initial phase, with this trend continuing a year later. The empirical results suggest that the service sector has a greater capacity for technological adjustment. In contrast, manufacturing workers maintained stable technology usage, indicating a slower, more gradual approach to technological upgrades.

Skill-level analysis reveals a clear adaptive advantage among high-skilled workers. Those concerned about AI displacement increased their computer and internet usage from 7.221 % before ChatGPT to 14.083 % after its launch. Their subjective technological well-being also improved, reflecting their ability to adapt and leverage AI as a skill advantage. In contrast, low-skilled workers showed only slight increases in technology usage, and their well-being remained low, underscoring their challenges in adapting to AI and the need for targeted skill training programs.

This study makes four key contributions. First, applying the Difference-in-Differences model captures workers' adaptation to AI technologies across different stages, addressing gaps in understanding AI's dynamic impact. Second, dividing the AI shock into two phases—before and during ChatGPT's launch and one year after—provides detailed insights into workers' behavioral and psychological responses to substitution pressures. Third, subsample analyses compare heterogeneous responses across industries (manufacturing and services) and skill levels (high- and low-skilled workers), offering empirical evidence to guide labor market adaptation strategies. Fourth, it introduces a novel theoretical lens by integrating behavioral economics and technology acceptance perspectives to explain the observed "fear → adaptation → divergence" pattern. This conceptual extension accounts for workers' psychological disequilibrium under AI disruption and the subsequent bifurcated adaptation paths shaped by skill level and sectoral digital readiness, offering a new framework for understanding AI-related behavioral change.

The structure of this paper is as follows. Section 2 reviews the literature, summarizing the impact of AI on the labor market and the relevant theoretical foundations. Section 3 describes the data sources. Section 4 outlines the theoretical model and empirical methodology. Section 5 presents the empirical results, focusing on workers' behavioral adjustments and changes in subjective technological well-being under the influence of AI technologies. Finally, Section 6 discusses the policy implications and concludes the study.

2. Literature review

The rapid advancement of AI and generative AI reshapes labor market structures and dynamics. These technologies redefine workforce distribution, alter skill demands, and influence productivity and innovation. However, their effects vary across industries and skill levels, creating opportunities and challenges. This section reviews existing literature on AI's role in boosting productivity and innovation, its disruptive impact on employment through technological substitution, and its varied effects across industries and regions. It also highlights this study's contributions in addressing key gaps in current research.

Furman and Seamans (2019) classify AI as a General-Purpose Technology (GPT) that improves resource allocation and drives firm-level innovation. The rise in AI patent filings reflects its growing integration into production systems. Damioli et al. (2023) find that AI-related patents boost employment growth in upstream sectors more than non-AI patents. Babina et al. (2024) further show that AI shortens product development cycles, enhances product quality, and accelerates innovation, leading to higher sales, employment growth, and increased market value. Notably, AI has been incredibly effective in generating high-skilled jobs, such as AI engineers and data scientists, shaping new employment opportunities.

This trend extends across healthcare, finance, education, and transportation industries. Sharma and Sehgal (2023) show that AI improves medical diagnostics, enhances financial risk management, strengthens student learning outcomes, and optimizes logistics. Nalwade and Misal (2024) find that AI restructures manufacturing by optimizing supply chains and improving quality control. Generative AI, like ChatGPT, has further broadened AI's applications. Chen et al. (2023) highlight that Generative AI creates new job roles and redefines skill requirements in traditional positions, reshaping the labor market.

However, research indicates a technological lag constrains AI's impact on firm productivity. Furman and Seamans (2019) attribute this delay to the time needed for commercialization, which depends on workforce training and infrastructure investments. Similar lags were observed with past General-Purpose Technologies, such as electrification and integrated circuits. Owolabi et al. (2024) add that while generative AI tools like ChatGPT enhance data analysis, they are limited in hypothesis testing and code execution, making them supportive rather than fully substitutive of human expertise. Babina et al. (2024) highlight that AI's benefits are concentrated in large enterprises, which leverage their data and resources more effectively. In contrast, small and medium-sized enterprises (SMEs) face resource constraints, limiting their AI adoption and widening market inequalities.

Recent evidence confirms that AI's impacts vary markedly by skill level. Brynjolfsson et al. (2025) provide micro-level findings consistent

with skill-biased technological change, showing that generative AI adoption yielded significant productivity gains with uneven distribution. A large firm's experiment found the most significant improvements accrued to less-experienced, lower-skilled workers, while the most experienced, high-skilled workers saw only marginal gains. This outcome suggests that AI tools can disseminate expert know-how to lower-skill employees, effectively complementing higher-skill work while automating routine tasks. Such findings align with the notion that AI tends to augment high-skilled workers' productivity but threatens lower-skilled, routine roles. The results provide directional support for these patterns: high-skill Taiwanese workers in the data adapted more quickly and benefited from AI-related changes, whereas low-skill workers struggled to keep up. However, given the data and design limitations (survey-based perceptions and no individual panel tracking), this evidence remains suggestive rather than conclusive. This study cautiously interprets the skill-differentiated effects of AI, acknowledging that the analysis cannot definitively establish causality even as it broadly comports with Brynjolfsson et al.'s empirical observations.

From a labor perspective, AI affects workers differently based on skill levels. Song (2024) finds that while AI boosts high-skilled workers' productivity, it increases structural unemployment risks for low-skilled workers by automating routine tasks and deepening labor market inequalities. Sharma and Sehgal (2023) further note that data bias and privacy concerns undermine trust in AI, complicating its broader implementation. Similarly, Owolabi et al. (2024) emphasize that limitations in AI model design and data coverage reduce flexibility in meeting specific contextual demands.

The rapid spread of AI technologies has intensified the substitution effect, placing significant pressure on low-skilled workers. Fatun and Paxour (2021) estimate that AI could replace 11 % of occupational skills in the short term, rising to 70 % in the medium to long term. This trend threatens low-skilled jobs, widens income inequality, and limits opportunities, as Acemoglu et al. (2022) noted. Similarly, Song (2024) highlights that automating routine tasks reduces demand for low-skilled labor, contributing to structural unemployment and labor market polarization. Sharma and Sehgal (2023) further point out that while AI reduces the need for routine jobs, it creates opportunities for high-skilled workers in creative, analytical, and decision-making roles.

However, skill transformation often outpaces workers' ability to adapt. Cramarenco et al. (2023) find that technostress negatively affects low-skilled workers' mental health and job satisfaction. Wang et al. (2025) report that robotics technology prolongs unemployment for low-skilled workers, particularly those in routine occupations. Wang et al. (2024) also highlight a "crowding-out effect," where digital technologies limit mobility for low-skilled workers but enhance cross-industry mobility for high-skilled workers. This effect is especially pronounced in manufacturing, where digitalization displaces workers more than in the service sector.

AI's labor substitution effects contrast with its role in job creation through technological innovation. Damioli et al. (2023) find that AI-driven product innovation supports employment growth in innovation-oriented firms. In contrast, Zheng et al. (2024) show that automation improves firms' economic performance but reduces demand for low-skilled labor, especially in high-pollution industries and regions with weak regulations.

AI's impact on the labor market reflects a "dual effect." On the one hand, AI-driven efficiency gains and job creation (Damioli et al., 2023; Sharma & Sehgal, 2023) benefit high-skilled workers. On the other hand, job displacement among low-skilled workers (Fatun & Paxour, 2021; Song, 2024) and the psychological burden of technological change (Cramarenco et al., 2023) deepen labor market inequalities. These challenges highlight the urgent need for targeted skill retraining programs and policy interventions, making them a priority for researchers and policymakers.

The rapid advancement of AI has had varied effects across industries and skill levels, reinforcing substitution for low-skilled workers while enhancing opportunities for high-skilled workers. Industry-specific impacts are evident. Wang et al. (2024) analyzed digitalization in China's manufacturing and service sectors, finding that manufacturing digitalization creates a crowding-out effect, limiting mobility for low-skilled workers while increasing cross-industry mobility for high-skilled workers. In contrast, digitalization in services stabilizes high-level positions but further restricts career opportunities for low-skilled workers.

Similarly, Wang et al. (2025) examined robotics in manufacturing and logistics, showing that low-skilled workers in routine jobs face more significant unemployment risks and prolonged joblessness due to automation. Zheng et al. (2024) found that automation reduces demand for low-skilled workers in high-pollution industries and weakly regulated regions. In contrast, high-tech industries use automation to boost productivity and expand labor demand, underscoring AI's uneven effects across industries.

Building on this, Wang and Lu (2025) provided evidence from Taiwan that AI displacement concerns vary by gender, age, education, and industry. Their findings highlight AI's complementary role in sectors focused on sustainable technologies, offering insights into AI's complex labor market effects.

AI's impact on skill levels is also pronounced. Song (2024) notes that AI enhances productivity and creates new job opportunities for high-skilled workers, such as data scientists, through technological innovation. In contrast, automation displaces low-skilled workers in routine tasks, increasing structural unemployment. Zheng et al. (2024) similarly emphasize that automation accelerates labor substitution with capital, particularly reducing low-skilled employment in low-tech industries.

Wang et al. (2024) stress that digitalization amplifies skill matching, helping high-skilled workers adapt to industrial shifts while creating mobility barriers for low-skilled workers. These challenges are most severe in manufacturing and logistics, where technological displacement is more pronounced (Wang et al., 2025). Meanwhile, high-tech industries leverage AI to enhance productivity, offering expanded opportunities for high-skilled workers (Zheng et al., 2024; Song, 2024).

AI has widened skill-based disparities, though its impact varies across industries. Low-skilled workers in manufacturing and logistics face significant substitution challenges (Wang et al., 2025; Zheng et al., 2024), while high-skilled workers in high-tech industries benefit from AI-driven innovation (Song, 2024; Zheng et al., 2024). To address this divide, future policies should focus on skill retraining for low-skilled workers (Wang et al., 2024; Song, 2024) while supporting further advancements for high-skilled workers. These efforts are crucial for balancing technological progress with workforce stability (Wang et al., 2024; Zheng et al., 2024).

The rise of generative AI has redefined human-machine collaboration, making it central to industrial and technological development. Unuriode et al. (2024) argue that this collaboration reduces technology's substitution effects by improving workers' adaptability and job stability. Similarly, Sonawane and Khandalkar (2023) emphasize that generative AI enhances high-performance tasks, complementing rather than replacing human creativity and ethical judgment. Owolabi et al. (2024) illustrate this with data analysis, showing that tools like ChatGPT provide analytical support but require human oversight to ensure accuracy and mitigate algorithmic bias, reinforcing AI's complementary role in professional settings.

Chen et al. (2023) examine ChatGPT's labor market impact by analyzing changing skill demands. Their findings show that generative AI adoption has significantly increased demand for ChatGPT-related skills, particularly in high-paying industries such as technology, product development, and marketing. They predict that nearly half of future job roles will require generative AI expertise, highlighting a strong link between skill proficiency and wage growth.

Research consistently affirms that generative AI is an assistive tool rather than a replacement for human labor. Sonawane and Khandalkar (2023) and Owolabi et al. (2024) stress AI's limitations in complex

decision-making and ethical reasoning, reinforcing the need for human involvement. Chen et al. (2023) further emphasize the rising demand for AI-related skills, underscoring the importance of workforce training.

While studies agree on generative AI's role, they focus on different aspects. Sonawane and Khandalkar (2023) highlight AI's technical limitations and the necessity of human creativity. Owolabi et al. (2024) examine its auxiliary role in professional applications. Chen et al. (2023) explore the increasing demand for AI-related skills and their impact on wages. Together, these findings suggest that optimal AI use lies in fostering human-AI collaboration to balance technological progress with workforce adaptability.

Existing research has extensively examined AI's impact on the labor market, focusing on technological substitution and skill transformation. However, the rise of generative AI, such as ChatGPT, presents new challenges and reveals critical research gaps.

First, most studies focus on AI's static effects, providing limited insight into the dynamic processes and long-term impacts of generative AI. While research has explored short-term substitution pressures on low-skilled workers, there is little analysis of how workers adapt behaviorally and psychologically over time (Chen et al., 2023; Wang et al., 2024).

Second, studies on generative AI's regional and economic impacts remain scarce. Existing research primarily focuses on developed economies, overlooking their adoption and effects in emerging markets. The role of policy interventions and regional characteristics in shaping workers' adaptability requires further exploration (Zheng et al., 2024).

Third, research on generative AI's ethical challenges and skill-shaping effects is still in its early stages. While studies highlight technological benefits and applications, few systematically examine workers' psychological adaptation and long-term behavioral changes. Additionally, policies to mitigate AI's adverse effects on low-skilled workers while maximizing benefits for high-skilled workers remain underdeveloped (Sonawane & Khandalkar, 2023; Owolabi et al., 2024).

This study addresses these gaps by analyzing official survey data from Taiwan. It examines the dynamic effects of generative AI during its initial adoption and stabilization phases, focusing on workers' behavioral adaptations and subjective well-being in response to technological change. Using a Difference-in-Differences model, it captures workers' adaptation processes under technological uncertainty. The study further explores heterogeneous impacts across industries (manufacturing and services) and skill levels (high to low), contributing to research on emerging economies and multi-level effects. Finally, integrating behavioral economics and the Technology Acceptance Model, it investigates the long-term psychological and behavioral effects of generative AI. By bridging these theoretical perspectives, the study extends conventional adoption models to account for workers' initial risk aversion and bounded rationality under uncertainty alongside TAM's focus on perceived usefulness and ease of use. In particular, this integrated framework highlights how psychological stress and perceived control evolve—from an initial anxiety response to mid-term adjustment as self-efficacy grows—ultimately influencing a diverging set of adaptation outcomes. Framing the observed fear–adaptation–divergence trajectory within this hybrid model provides a novel conceptual contribution, adding nuance to prior technology acceptance theories. The findings provide targeted policy recommendations for skill retraining and technological adaptation, offering valuable insights for academia and policymakers.

3. Data sources

This study utilizes data from Taiwan's Digital Development Survey (National Development Council, 2022) and Digital Opportunity Survey (Ministry of Digital Affairs, 2022; 2023), both provided by the Taiwan government. These nationally representative datasets include indicators of digital technology usage, internet applications, digital practices, and attitudes toward technological development. The analysis focuses on three key time points: 2020 (before ChatGPT's emergence), 2022 (the year ChatGPT was launched), and 2023 (one year after its introduction). Due to the large-scale COVID-19 outbreak in Taiwan in 2021, the survey was suspended that year, excluding 2021 data. Notably, each year's survey is a repeated cross-sectional sample rather than a panel of the same individuals. Thus, we cannot directly track specific workers over time. As a result, any changes observed across 2020, 2022, and 2023 reflect overall trends at the group level and cannot measure an individual's change in job typology or task composition over time. Using data collected multiple times, the study examines workers' concerns about AI replacing job functions, behavioral responses, and changes in subjective technological well-being over time.

The sample design captures Taiwan's segmented labor market. National statistics indicate that about 60 % of Taiwanese workers are employed in services and around 35 % in manufacturing, aligning with our broad division of survey respondents into the service and manufacturing sectors. The survey's coverage spans this sectoral split and various occupations. Notably, Taiwan's workforce is broadly educated (with high secondary and tertiary education rates), but significant skill gaps persist. Defining four skill tiers – high to low – is therefore pertinent, as Taiwan exhibits features of a polarized labor market where many high-skill jobs co-exist with a vulnerable lower-skill segment.

The primary independent variable is workers' level of concern about AI replacing their job tasks (often labeled $AIreplace_i$), which is measured as an ordinal category with four levels: "very worried," "somewhat worried," "not very worried," and "not worried at all." In the empirical model, workers who are "not worried at all" about AI replacement are the baseline (control group), indicating no perceived displacement risk. At the same time, the other three categories represent increasing degrees of concern about AI-driven job loss. The dependent variables are two key outcomes capturing technology use and well-being. First, computer/internet use in job tasks is measured by whether a worker's role requires using computers and the internet. This is a yes/no item in the survey, which we code as a binary variable (1 if the individual uses computers/internet in their job, 0 otherwise). This "job-required computer and internet use" indicator is analyzed at the individual level and summarized as a proportion of workers using digital tools for group comparisons. Second, subjective technological well-being refers to workers' self-reported comfort and confidence with workplace technology. This is an ordinal self-rating (from low to high) of technological well-being, where higher values indicate greater ease and positivity in dealing with new technologies. This measure captures the psychological impact of technological change on the worker (for example, a higher score suggests the worker feels more adept and less stressed with technology in their job). Both dependent variables are derived from individual survey responses. Due to the survey design, we cannot observe the frequency or intensity of digital tool use – only whether such tools are used in the respondent's job.

To assess heterogeneous impacts, the study differentiates industries and skill levels. Industry classifications follow the definitions from the Ministry of Digital Affairs and the Directorate General of Budget, Accounting, and Statistics, dividing industries into manufacturing and services to analyze behavioral differences across sectors. Skill levels are categorized into four tiers—high-skill, medium-high-skill, medium-skill, and low-skill—capturing adaptability variations based on job roles' technical demands. For example, high-skill occupations (e.g., software engineers or senior managers) require advanced expertise, medium-high-skill jobs (e.g., technicians or associate professionals) involve considerable technical or administrative skill, medium-skill jobs (e.g., office clerks or administrative assistants) entail moderate skill requirements, and low-skill jobs (e.g., assembly-line operators or retail service workers) comprise routine tasks with minimal specialized skills.

Using a Difference-in-Differences model, the study analyzes workers' adjustment processes before and after ChatGPT's emergence. It examines how AI-related job displacement concerns influence workers'

behavior and technological well-being, considering differences across industries and skill levels. The findings provide empirical evidence on AI's labor market effects and contribute to understanding how workers adapt behaviorally and psychologically to generative AI.

4. Empirical model

This study empirically examines the causal relationship between worker behavioral adjustments and changes in subjective technological well-being driven by AI. The analysis employs a Difference-in-Differences (DID) model. It uses cross-period comparisons on repeated cross-sectional survey data to capture AI's dynamic effects during its initial introduction and the following year. Since the data do not follow the same individuals over time, the empirical model includes controls for workers' demographic and job-related characteristics to mitigate differences in sample composition across the three survey waves. The study establishes its theoretical foundation and develops an empirical framework to support the investigation.

As an emerging transformative technology, AI often drives firms to restructure production processes and operations. In response to technological uncertainty, workers must proactively adapt their skills and job roles to reduce potential disruptions and maintain labor market stability.

In the model framework of this study, it is assumed that the process of AI adoption experienced by worker i can be divided into three consecutive stages: t = 0,1,2. In the first stage (t = 0), before the launch of ChatGPT, workers operate in a production environment unaffected by this emerging technology. Consequently, workers determine their skill combinations and task allocations $x_i(0)$ based solely on existing technological conditions and production requirements while incurring the corresponding costs of skill investment and adjustment $c(x_i(0))$.

In the second stage (t = 1), with ChatGPT officially launched in 2022, it integrates into existing production systems ($T(1)$). At this stage, workers face the partial penetration of the new technology and must make moderate reallocations of their tasks and skills ($x_i(1)$). As these adjustments are incremental and occur within the bounds of the existing production framework, the associated skill adjustment costs are likely to be relatively low and may follow the trajectory of prior skill investments.

In the third stage (t = 2), one-year after the introduction of ChatGPT, the new technology becomes profoundly integrated and stably applied across various task workflows. At this point, the diffusion of the technology reaches maturity, and its impacts on task composition, skill requirements, and production modes become more evident and stable. Confronted with this stable yet profound transformation of the technological ecosystem, workers must undertake comprehensive adjustments to their skill combinations and task allocations ($x_i(2)$). They must also reassess their skill investment strategies and task arrangements within a new production equilibrium context. The contents of these three phases are summarized in Table 1.

Table 1
Timeline of technology adoption and workers' decision-making process.

Period	$t = 0$ (2020)	$t = 1$ (2022)	$t = 2$ (2023)
Phase	No ChatGPT	ChatGPT Public Launch	ChatGPT and Worker Technology Integration
Technological Status	Traditional Production Technologies $T(0) = 0$	Partial Adoption of AI Tools $T(1) = 1$	Deep Integration of AI Technologies $T(2) = 1$
Worker Decisions	Initial skill and task allocation based on traditional technologies and AI influences $x_i(0)$	Minor adjustments to skills and task allocation based on existing skills $x_i(1)$	Re-optimization of skills and task allocation under a mature AI production ecosystem $x_i(2)$

During the process of AI adoption, the worker's objective is to maximize their intertemporal expected utility, expressed as:

$$\underset{x_i(0),x_i(1),x_i(2)}{Max} E\left[U_i(0) + \alpha U_i(1) + \alpha^2 U_i(2)\right] \tag{1}$$

This study hypothesizes that workers' overall utility derives from two primary dimensions: tangible compensation (wage and benefits) and subjective technological well-being. To better delineate workers' decision-making under different technology states, this study defines $W_i(t)$ as the level of tangible rewards, encompassing wages and workplace benefits, experienced by worker i in period t.

$$W_i(t) = f(x_i(t), T(t)) \tag{2}$$

$x_i(t)$ represents the skill configuration and task combination chosen by worker ii in period tt, while $T(t)$ serves as an indicator of the technological state during that period. Specifically, $T(0) = 0$ denotes the traditional production technology state before the introduction of ChatGPT. When $T(1) = T(2) = 1$, it signifies the integration of ChatGPT into labor processes, with the maturity of its application at $t = 2$ being higher than at $t = 1$.

Let $S_i(t)$ denote the subjective technological well-being experienced by worker ii in period t. This well-being arises from the perceived impacts of new artificial intelligence technologies, such as ChatGPT, on job content, skill requirements, and working conditions.

$$S_i(t) = S(A_i, x_i(t), T(t)) \tag{3}$$

Here, $A_i \in \{1, 2, 3, 4\}$ represents worker i's concern regarding AI replacing their job. Smaller values indicate more significant concern, with $A_i = 4$ denoting workers who are completely unconcerned about being replaced by AI, forming the control group. Meanwhile, $A_i = 1, 2, 3$ represent varying degrees of perceived AI-related job displacement risk, corresponding to the treatment group: $A_i = 1$ indicates "very worried," $A_i = 2$ indicates "somewhat worried," and $A_i = 3$ indicates "not very worried." Based on these definitions, the worker's overall utility function can be expressed as the weighted sum of tangible compensation and technological well-being:

$$U_i(t) = W_i(t) + \beta S_i(t) \tag{4}$$

In the initial state of ChatGPT integration ($t = 0$), workers base their skill configuration choices, $x_i(0)$, on traditional technologies, incurring corresponding costs. Their utility at this stage is expressed as:

$$U_i(0) = f(x_i(0), 0) + \beta S(A_i, x_i(0), 0) - c(x_i(0)) \tag{5}$$

When ChatGPT is publicly launched ($t = 1$), and during its diffusion and stabilization ($t = 2$), workers adjust their skill configurations and job assignments based on the conditions of the new technology. Their utility in these periods can be expressed as:

$$U_i(1) = f(x_i(1), 1) + \beta S(A_i, x_i(1), 1) \tag{6}$$

$$U_i(2) = f(x_i(2), 1) + \beta S(A_i, x_i(2), 1) \tag{7}$$

Due to differing levels of concern among workers regarding AI job replacement, their decision-making at $t = 0$ and $t = 1$ will vary. Through early and mid-stage adjustments to skill configurations and task assignments, distinct levels of output and well-being may emerge at $t = 2$. Given that most workers exhibit risk-averse attitudes, those concerned about AI replacement are likely to proactively shift toward tasks complementary to AI. For example, they may increase the use of computers and the internet in their work processes to sustain higher levels of output and well-being after ChatGPT's launch.

These differences manifest to varying degrees across two-time intervals. From $t = 0$ to $t = 1$, workers with a more significant concern may experience early-stage changes in job characteristics and well-being during the initial adoption of the technology. However, from $t = 1$ to $t = 2$, these workers are expected to undergo more significant behavioral

adjustments, leading to more excellent stability or pronounced differentiation in job characteristics and well-being outcomes. These trends reflect the ongoing divergence influenced by AI integration.

Finally, when workers aim to maximize their utility $U_i(t)$ in period t, the first-order optimization condition for $x_i(t)$ can be derived. Solving this condition yields the optimal skill configuration and task allocation for workers in period tt that maximizes their utility, expressed as:

$$x_i^*(t) = \underset{x_i(t)}{\operatorname{argmax}}\, U_i(t) \tag{8}$$

The mathematical model above reveals workers' optimal behavioral rules for adjusting their skills and task configurations under varying technological states. However, endogenous variables in the theoretical model, such as utility and optimal skill combinations, are not directly observable in empirical data. Therefore, this study maps the theoretical conclusions to observable labor market behavior indicators, such as the proportion of computer and internet use required for work and subjective technological well-being.

The theoretical model treats wages as a component of workers' utility, but this factor is included only conceptually and is not incorporated into the empirical analysis. The survey data do not provide detailed wage or income information for the periods in question, so the wage variable cannot be analyzed in the Difference-in-Differences models. Instead, the empirical analysis centers on the observable outcomes of technology use and well-being. Similarly, while the model posits that workers incur skill adjustment or learning costs when adapting to new AI technology, such learning costs are excluded from the empirical specifications due to their latent nature and data limitations. This study lacks direct measures (e.g., training hours or expenditures) to quantify individual learning costs, making them impractical to include. The omission of wages and learning costs in the regressions is theoretically justified because these variables, though important for utility, are not directly observable; their effects are instead assumed to be reflected indirectly in the workers' behavioral adjustments and subjective well-being outcomes that this study does measure.

An econometric model capable of empirical identification is required to test the theoretical predictions. This study employs two samples: the first spans the period before ChatGPT's emergence (2020) to its public launch year (2022), while the second covers the period from the launch year (2022) to one-year post-launch. Considering the research objectives, sample structure, and theoretical model, this study adopts the Difference-in-Differences (DID) method to combine temporal differences (ChatGPT development timeline) and group differences (workers' concerns about job displacement) to capture the causal effects of AI technology adoption.

The degree of workers' concern about AI job replacement across different years can be expressed as:

$$AIreplace_i = \begin{cases} 1 & \textit{very worried} \\ 2 & \textit{somewhat worried} \\ 3 & \textit{not very worried} \\ 4 & \textit{not worried at all} \end{cases} \quad if \tag{9}$$

In the actual estimation, $AIreplace_i = 4$ is the control group sample. For analyzing the impact before and after the launch of ChatGPT in Sample 1, the DID model can be expressed as follows:

$$\begin{aligned} Y_{it} = \gamma_0 &+ \sum_{a=1}^{3} \gamma_{1a}(AIreplace_i = a) + \gamma_2 Post_t^{0\to1} \\ &+ \sum_{a=1}^{3} \gamma_{3a}\left((AIreplace_i = a) \times Post_t^{0\to1}\right) + \varepsilon_i \end{aligned} \tag{10}$$

Y_{it} can represent two proxy variables for evaluation: the proportion of tasks requiring computers and the internet to complete work and subjective technological well-being. The variable $Post_t^{0\to1} = 1$ indicates the year 2022, marking the public launch of ChatGPT.

$$\begin{aligned} Y_{it} = \lambda_0 &+ \sum_{a=1}^{3} \lambda_{1a}(AIreplace_i = a) + \lambda_2 Post_t^{1\to2} \\ &+ \sum_{a=1}^{3} \lambda_{3a}\left((AIreplace_i = a) \times Post_t^{1\to2}\right) + e_i \end{aligned} \tag{11}$$

$Post_t^{1\to2} = 1$ represents 2023, one-year after ChatGPT's launch. In equations (9) and (10), the primary focus is on the coefficients γ_3 and λ_3, which capture the additional differential changes from the pre-launch to launch period and from launch to one-year post-launch. These coefficients reflect the two-stage adjustment effects in workers' adaptation to the impacts of AI technology relative to the control group of workers who are completely unconcerned about AI replacing their jobs.

In the empirical Difference-in-Differences setup, the "Post" dummy variable denotes the period after ChatGPT's introduction (with 1 for the post-introduction period and 0 for the pre-introduction period). Specifically, in analyses comparing 2020 (pre-ChatGPT) to 2022, the "Post" period corresponds to 2022 – the ChatGPT launch year. In analyses comparing 2022 to 2023, the "Post" period corresponds to 2023 – one year after ChatGPT's launch.

Additionally, as workers' needs to integrate AI into their tasks may vary significantly depending on the industry and job type, the empirical analysis accounts for such heterogeneity. This study classifies industries into manufacturing and services categories based on the Standard Industrial Classification (SIC) by Taiwan's Directorate-General of Budget, Accounting, and Statistics (DGBAS). Furthermore, job types are categorized into four skill levels, ranked from high to low, following the Standard Occupational Classification (SOC). These classifications allow for a detailed subsample analysis across different industries and occupational skill levels.

The academic contribution of this two-stage DID analysis lies in its theory-driven approach, which translates workers' subjective perceptions of AI replacement risks into operationalized empirical groupings. By comparing outcomes across distinct periods, the study highlights the adaptation processes during the initial phase of new technology adoption and its subsequent deepening stage. This dynamic perspective underscores that technological impacts are not one-time events but evolve, with corresponding adjustments in workers' behavior and psychological states. The findings enrich the understanding of the labor market's dynamic adaptation to the rapid diffusion of AI technologies, offering a more nuanced depiction of this transitional process.

5. Empirical results

This study applies the DID method, using nationally representative survey data to conduct a dynamic cross-period analysis. It examines full-time workers in Taiwan and their perceptions of AI-driven job displacement risks across three key periods: before ChatGPT's emergence, its public launch, and one-year post-launch. The analysis focuses on how these perceptions influence the proportion of tasks requiring computer and internet use and subjective technological well-being. The following sections present a detailed discussion of the empirical results.

5.1. Analysis of the impact of workers' perceptions of AI job replacement risks

The DID analysis results in Table 2 show that ChatGPT's introduction significantly impacted workers' digital technology adoption and subjective technological well-being. Before ChatGPT's official launch, the proportion of workers using computers and the internet increased notably. This proportion rose by 10.175 % among highly concerned workers, while moderately and slightly concerned workers saw increases of 7.616 % and 6.328 %, respectively. These findings indicate that concerns about AI-driven job displacement prompted workers to adopt digital tools to mitigate risks and improve their competitiveness.

Table 2
Empirical results of the impact analysis on workers' concerns about job displacement by AI.

Difference-in-Differences	Job Requiring Computer and Internet Use Rate		Subjective Well-Being in Tech Trends	
	Pre-Post ChatGPT Launch Year	ChatGPT Launch to One Year Later	Pre-Post ChatGPT Launch Year	ChatGPT Launch to One Year Later
Workers' Concern on AI-Induced Job Loss				
AIdisplace = 1	10.175*** (5.59)	2.989 (1.52)	−0.232*** (−2.69)	−0.429*** (−4.67)
AIdisplace = 2	7.616*** (5.17)	7.762*** (5.07)	−0.034 (−0.49)	−0.234*** (−3.28)
AIdisplace = 3	6.328*** (5.06)	5.418*** (4.27)	0.043 (0.73)	−0.020 (−0.35)
Post	0.638 (0.45)	−1.557 (−1.07)	0.238*** (3.50)	−0.054 (−0.80)
Interaction Terms				
AIdisplace = 1 × Post	−7.186*** (−2.68)	9.606*** (3.54)	−0.197 (−1.55)	0.345*** (2.72)
AIdisplace = 2 × Post	0.146 (0.07)	5.071** (2.41)	−0.199** (−1.99)	0.210** (2.14)
AIdisplace = 3 × Post	−0.909 (−0.51)	3.689** (2.06)	−0.063 (−0.76)	0.063 (0.76)
F value	10.71*** (0.00)	19.29*** (0.00)	9.41*** (0.00)	5.49*** (0.00)
N	12425	12613	12425	12613

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. ***, ** and * represent statistical significance at the 1 %, 5 % & 10 %, respectively. d. 'Post' equals 1 in the post-ChatGPT period and 0 otherwise. In the 2020–2022 comparison, Post = 1 denotes 2022 (ChatGPT launch year); in the 2022–2023 comparison, Post = 1 denotes 2023 (one year after launch).

After ChatGPT's launch, highly concerned workers' reliance on digital technologies dropped significantly to 2.989 %, losing statistical significance. This decline may indicate increased technological pressure or role adjustments toward non-technical tasks, reducing technology usage. However, the marginal effects analysis in Table 3 shows a rebound in digital tool reliance, rising from 52.909 % at launch to 60.958 % one year later. This recovery suggests long-term adaptation, likely driven by expanded enterprise-led digital skills training and greater acceptance of new technologies.

Table 3
Empirical results of marginal effects analysis under different perceptions of AI job displacement concerns.

Marginal Effects	Job Requiring Computer and Internet Use Rate		Subjective Well-Being in Tech Trends	
	Pre-Post ChatGPT Launch Year	ChatGPT Launch to One Year Later	Pre-Post ChatGPT Launch Year	ChatGPT Launch to One Year Later
AIdisplace = 1, Post = 0	59.456	52.909	6.480	6.520
AIdisplace = 1, Post = 1	52.909	60.958	6.520	6.811
AIdisplace = 2, Post = 0	56.897	57.682	6.678	6.716
AIdisplace = 2, Post = 1	57.682	61.197	6.716	6.872
AIdisplace = 3, Post = 0	55.609	55.338	6.755	6.929
AIdisplace = 3, Post = 1	55.338	57.470	6.929	6.938
AIdisplace = 4, Post = 0	49.281	49.919	6.712	6.950
AIdisplace = 4, Post = 1	49.919	48.362	6.950	6.896

Note: The values in the table are all statistically significant at the 1 % level, indicating that technological dependence and subjective well-being aligned with technological trends for specific labor groups are highly stable and representative rather than random or coincidental outcomes.

Moderately concerned workers showed stable and gradual adaptation, with their reliance on digital technologies increasing from 57.682 % to 61.197 % one year after ChatGPT's launch. In contrast, workers with low or no concerns about AI replacement exhibited stable or slightly declining digital tool usage, suggesting minimal behavioral change and lower sensitivity to technological advancements.

Regarding subjective technological well-being, highly concerned workers experienced a decline to 6.52 during ChatGPT's launch but rebounded significantly to 6.811 a year later, reflecting the positive effects of long-term adaptation. Moderately concerned workers followed a similar trend, with their well-being index increasing from 6.716 to 6.872. Workers with low or no concerns showed minor changes, while the unconcerned group experienced a slight decline, suggesting limited psychological effects from ChatGPT's introduction.

Interaction term analysis revealed dynamic changes across concern levels. Before ChatGPT's launch, the interaction term was negative for highly concerned workers, indicating uncertainty and psychological pressure. Over time, this shifted to positive, signaling gradual adaptation and recovery in technology use and psychological well-being. Similarly, moderately concerned workers transitioned from unfavorable to favorable interaction terms, reflecting improved adaptability and attitudes toward technology. These patterns align with the Technology Acceptance Model, which suggests that perceived usefulness and ease of use drive technology adoption, facilitating behavioral and psychological adaptation.

These findings highlight ChatGPT's heterogeneous impacts on workers' behaviors and well-being. The dynamic adjustments in digital technology use and subjective well-being underscore the interaction between psychological expectations and reality. The results emphasize the importance of external support, such as corporate training and internal psychological adaptation, in mitigating AI-induced disruptions. They provide empirical evidence for designing targeted policies that enhance workers' technical skills and psychological resilience in response to rapid technological change.

5.2. Analysis of AI replacement concerns across different industries

This study employs a DID model to analyze the impact of AI technologies on workers' digital technology reliance and subjective technological well-being across the manufacturing and service sectors. The results indicate significant industry-level differences in response to AI-related challenges.

As shown in Table 4, before ChatGPT's launch, the proportion of manufacturing workers using computers and the Internet for work was 9.053 %, 5.406 %, and 4.508 % higher among highly, moderately, and slightly concerned workers, respectively, compared to unconcerned workers. In the service sector, these increases were 10.496 %, 8.487 %, and 7.042 %, with higher statistical significance. These findings highlight the service sector's greater sensitivity to technological demands, reflecting its reliance on software applications and internet tools. In contrast, the manufacturing sector's relatively higher baseline digitalization and stable production processes resulted in more limited short-term adjustments.

Following ChatGPT's launch, the gap between the two sectors widened. In manufacturing, highly concerned workers relied more on digital tools than their unconcerned counterparts. However, growth was limited, and interaction terms remained insignificant. This suggests that hardware upgrades, process modifications, and quality control requirements constrain technology adoption in manufacturing. In contrast, the service sector showed greater adaptability. Reliance on digital tools initially dropped from 62 % to 54.1 % among highly concerned workers but rebounded to 65.1 % within a year. Interaction terms shifted from negative to positive and became statistically

Table 4
Empirical results of the impact analysis on workers' concerns about job displacement by AI in manufacturing and service industries.

Difference-in-Differences	Job Requiring Computer and Internet Use Rate				Subjective Well-Being in Tech Trends			
	Pre-Post ChatGPT Launch Year		ChatGPT Launch to One Year Later		Pre-Post ChatGPT Launch Year		ChatGPT Launch to One Year Later	
	Mfg.	Svc.	Mfg.	Svc.	Mfg.	Svc.	Mfg.	Svc.
Workers' Concern on AI-Induced Job Loss (AIdisplace)								
AIdisplace = 1	9.053*** (2.70)	10.496*** (4.87)	5.241 (1.58)	1.844 (0.76)	−0.102 (−0.63)	−0.290*** (−2.88)	−0.324** (−2.04)	−0.483*** (−4.30)
AIdisplace = 2	5.406** (2.01)	8.487*** (4.85)	6.838*** (2.57)	8.130*** (4.37)	−0.192 (−1.46)	0.035 (0.43)	−0.192 (−1.51)	−0.261*** (−3.05)
AIdisplace = 3	4.508** (1.98)	7.042*** (4.73)	7.612*** (3.44)	4.072*** (2.65)	−0.091 (−0.82)	0.102 (1.48)	0.009 (0.09)	−0.047 (−0.67)
Post	1.229 (0.49)	0.705 (0.41)	−2.875 (−1.15)	−0.908 (−0.51)	0.101 (0.82)	0.318*** (3.93)	−0.124 (−1.04)	−0.019 (−0.23)
Interaction Terms								
AIdisplace = 1 × Post	−3.812 (−0.81)	−8.652*** (−2.66)	5.105 (1.10)	11.980*** (3.59)	−0.221 (−0.96)	−0.193 (−1.27)	0.458** (2.07)	0.281* (1.83)
AIdisplace = 2 × Post	1.432 (0.38)	−0.356 (−0.14)	2.594 (0.70)	5.973** (2.34)	0.0008 (0.00)	−0.297** (−2.49)	0.169 (0.96)	0.220* (1.88)
AIdisplace = 3 × Post	3.104 (0.98)	−2.969 (−1.39)	1.938 (0.62)	4.408** (2.04)	0.101 (0.66)	−0.150 (−1.50)	0.206 (1.37)	−0.009 (−0.10)
F value	3.59*** (0.00)	8.62*** (0.00)	5.03*** (0.00)	16.59*** (0.00)	1.92*** (0.06)	9.18*** (0.00)	2.09** (0.04)	4.48*** (0.00)
N	4076	8349	4248	8365	4076	8349	4248	8365

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. ***, ** and * represent statistical significance at the 1 %, 5 % & 10 %, respectively. d. 'Post' equals 1 in the post-ChatGPT period and 0 otherwise. In the 2020–2022 comparison, Post = 1 denotes 2022 (ChatGPT launch year); in the 2022–2023 comparison, Post = 1 denotes 2023 (one year after launch).

significant, indicating that skill enhancement and adaptation enabled service sector workers to overcome initial uncertainties and adopt technology more effectively. Moderately and slightly concerned workers also showed steady increases, reflecting a broader digital transformation in the sector.

Subjective technological well-being followed distinct patterns across sectors. In manufacturing, differences across concern levels were insignificant before and during ChatGPT's launch, suggesting stable psychological expectations. However, highly concerned workers saw their well-being recover from 6.442 during the launch year to 6.776 a year later, indicating reduced psychological pressure and renewed confidence. In the service sector, highly concerned workers experienced a sharper trajectory, with well-being initially declining from 6.443 to 6.568 but then rebounding significantly to 6.830. This recovery closed the gap with other groups and, in some cases, exceeded them, suggesting that skill enhancement and psychological adjustments supported long-term adaptation to AI integration.

Table 5's marginal effects analysis quantifies these trends. In manufacturing, highly concerned workers' reliance on digital tools declined from 53.6 % before ChatGPT to 51.0 % at launch but recovered to 53.3 % within a year. Meanwhile, the unconcerned group's usage fell from 45.8 % to 42.9 %, possibly due to a lack of adaptation pressure. The service sector exhibited more dynamic changes. Highly concerned, workers' digital reliance dropped from 62 % to 54.1 % at launch but surged to 65.1 % within a year. Similarly, their well-being declined initially but rebounded from 6.443 to 6.830, reinforcing the success of their adaptation and psychological recovery.

The DID and marginal effects analyses indicate better stability in the manufacturing sector, reflecting its slower pace of technological adaptation. In contrast, the service sector exhibited higher dynamism and adaptability. These differences underscore the role of industry characteristics in shaping responses to technological change. Policymakers should implement targeted measures to support more extended adjustment periods in manufacturing, such as internal training programs to enhance worker adaptability. For the service sector, efforts should focus on reducing short-term psychological pressures and providing sustained support for technology integration to improve adaptation efficiency.

5.3. Analysis of AI replacement concerns across different Worker skill levels

This study explores how workers at different skill levels responded to ChatGPT's introduction, focusing on digital technology adoption and subjective technological well-being. The empirical results in Tables 6 and 7 reveal significant differences in adaptability, with high-skill workers exhibiting the most potent ability to adjust.

Table 5
Empirical results of marginal effects analysis under different perceptions of AI job displacement concerns in manufacturing and service industries.

Marginal Effects	Job Requiring Computer and Internet Use Rate				Subjective Well-Being in Tech Trends			
	Pre-Post ChatGPT Launch Year		ChatGPT Launch to One Year Later		Pre-Post ChatGPT Launch Year		ChatGPT Launch to One Year Later	
	Mfg.	Svc.	Mfg.	Svc.	Mfg.	Svc.	Mfg.	Svc.
AIdisplace = 1, Post = 0	53.621	62.008	51.038	54.061	6.563	6.443	6.442	6.568
AIdisplace = 1, Post = 1	51.038	54.061	53.267	65.133	6.442	6.568	6.776	6.830
AIdisplace = 2, Post = 0	49.973	60.000	52.635	60.348	6.473	6.769	6.574	6.790
AIdisplace = 2, Post = 1	52.635	60.348	52.353	65.413	6.574	6.790	6.619	6.992
AIdisplace = 3, Post = 0	49.076	58.554	53.409	56.290	6.574	6.836	6.777	7.004
AIdisplace = 3, Post = 1	53.409	56.290	52.472	59.790	6.777	7.004	6.858	6.976
AIdisplace = 4, Post = 0	44.567	51.512	45.796	52.217	6.665	6.734	6.767	7.052
AIdisplace = 4, Post = 1	45.796	52.217	42.920	51.308	6.767	7.052	6.642	7.033

Note: The values in the table are all statistically significant at the 1 % level, indicating that technological dependence and subjective well-being aligned with technological trends for specific labor groups are highly stable and representative rather than random or coincidental outcomes.

Table 6
Empirical results of the impact analysis on workers' concerns about job displacement by AI across different skill levels.

Difference-in-Differences	Job Requiring Computer and Internet Use Rate								Subjective Well-Being in Tech Trends							
	Pre-Post ChatGPT Launch Year				ChatGPT Launch to One Year Later				Pre-Post ChatGPT Launch Year				ChatGPT Launch to One Year Later			
	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor
Workers' Concern on AI-Induced Job Loss (AIdisplace)																
AIdisplace = 1	7.221***	14.083***	16.293***	16.390***	0.515	−0.080	6.624**	14.593***	−0.498***	−0.435**	0.127	0.775**	−0.668***	−0.358	−0.225*	−0.170
	(2.65)	(2.88)	(6.41)	(3.62)	(0.18)	(−0.02)	(2.45)	(2.92)	(−3.64)	(−2.00)	(1.00)	(2.15)	(−4.47)	(−1.53)	(−1.71)	(−0.44)
AIdisplace = 2	3.542*	8.480**	12.914***	12.377***	2.354	17.951***	11.743***	7.565**	−0.251**	−0.068	0.174	0.708**	−0.413***	−0.383**	0.021	−0.131
	(1.74)	(2.20)	(6.04)	(3.35)	(1.07)	(4.53)	(5.41)	(2.04)	(−2.45)	(−0.40)	(1.63)	(2.40)	(−3.68)	(−2.16)	(0.20)	(−0.45)
AIdisplace = 3	1.926	7.730**	11.961***	5.139*	−1.259	10.965***	8.688***	4.613 (1.46)	−0.166**	0.013	0.269***	0.388	−0.252***	−0.009	0.179**	−0.049
	(1.19)	(2.31)	(6.36)	(1.65)	(−0.73)	(3.31)	(4.72)		(−2.04)	(0.09)	(2.87)	(1.57)	(−2.88)	(−0.07)	(2.01)	(−0.20)
Post	4.085**	−1.670	2.760 (1.30)	2.836 (0.82)	−1.370	3.477 (0.92)	−3.463*	−1.612	0.216**	0.307*	0.339***	0.726***	−0.077	−0.071	−0.048	−0.120
	(2.13)	(−0.44)			(−0.70)		(−1.64)	(−0.46)	(2.25)	(1.81)	(3.19)	(2.64)	(−0.78)	(−0.43)	(−0.47)	(−0.44)
Interaction Terms																
AIdisplace = 1 × Post	−6.705*	−14.163**	−9.669***	−1.797	9.406**	16.109**	12.377***	1.138 (0.17)	−0.169	0.077	−0.353*	−0.945*	0.367*	0.093	0.391**	0.790
	(−1.65)	(−1.98)	(−2.61)	(−0.27)	(2.37)	(2.13)	(3.30)		(−0.83)	(0.24)	(−1.91)	(1.76)	(1.81)	(0.28)	(2.15)	(1.49)
AIdisplace = 2 × Post	−1.187	9.470*	−1.171	−4.812	4.637	−5.887	6.639**	6.178 (1.16)	−0.161	−0.315	−0.153	−0.840**	0.180 (1.19)	0.232	0.195	0.490
	(−0.39)	(1.72)	(−0.39)	(−0.92)	(−1.57)	(−1.05)	(2.22)		(−1.05)	(−1.28)	(−1.01)	(−2.02)		(0.93)	(1.34)	(1.18)
AIdisplace = 3 × Post	−3.186	3.235 (0.69)	−3.273	−0.526	5.615**	−6.837	5.856**	−1.030	−0.085	−0.023	−0.089	−0.438	0.230*	−0.140	0.009	0.218
	(−1.32)		(−1.25)	(−0.12)	(2.32)	(−1.48)	(2.25)	(−0.23)	(−0.71)	(−0.11)	(−0.68)	(−1.25)	(1.87)	(−0.68)	(0.07)	(0.62)
F value	2.07***	5.15***	13.36***	4.18***	4.70***	5.80***	19.89***	4.40***	6.85***	3.21***	5.91***	1.63 (0.12)	5.61***	1.53 (0.15)	2.76***	0.72
	(0.04)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)		(0.00)		(0.00)	(0.65)
N	4103	1536	6387	1102	3971	1561	6603	1073	4103	1536	6387	1102	3971	1561	6603	1073

a. Values in parentheses represent t-values. b. All regression models include a constant term. c. ***, ** and * represent statistical significance at the 1 %, 5 % & 10 %, respectively. d. 'Post' equals 1 in the post-ChatGPT period and 0 otherwise. In the 2020–2022 comparison, Post = 1 denotes 2022 (ChatGPT launch year); in the 2022–2023 comparison, Post = 1 denotes 2023 (one year after launch).

Among high-skill workers, the highly concerned group significantly increased their computer and internet usage from 7.221 % before ChatGPT to 14.083 % at launch and further to 16.109 % one year later. Marginal effects dropped slightly from 73.084 % to 70.464 % at launch but rebounded to 78.500 % after a year. Their well-being improved notably from 6.769 to 7.106, suggesting a strong link between rapid technology adoption and psychological recovery. In contrast, the unconcerned group showed limited growth, with usage rates increasing from 4.085 % to 9.406 %, while marginal effects peaked at 69.948 % before slightly declining to 68.578 % a year later.

Mid-high-skill workers adapted more gradually. The highly concerned group's usage rates rose from 8.480 % to 17.951 %, stabilizing at 16.109 % after a year. Marginal effects declined from 70.212 % to 54.378 % at launch but recovered to 73.965 % over time. Their subjective well-being improved steadily from 6.525 to 6.931, indicating successful adaptation despite initial psychological pressure. The unconcerned group exhibited weaker growth, with usage rates increasing marginally from −1.670 % to 3.477 %, while well-being rose modestly from 6.168 to 6.459.

Mid-skill workers showed less pronounced changes. The highly concerned group's usage rates dropped from 16.293 % to 11.743 %, with marginal effects falling from 51.749 % to 44.840 % before recovering slightly to 53.754 %. Their well-being improved modestly from 6.295 to 6.625, while the unconcerned group saw a slight increase from 6.168 to 6.459. These results suggest mid-skill workers had limited adaptability, with some potential for improvement among the highly concerned.

Low-skill workers exhibited the weakest performance in both technology adoption and well-being. The highly concerned group's usage rates declined slightly from 16.390 % to 14.593 %, while marginal effects saw only a modest increase from 24.918 % to 25.483 %. The unconcerned group experienced a decline in usage rates from 2.836 % to −1.612 %, with marginal effects rising from 8.527 % to 11.364 % before falling again to 9.751 %. Well-being changes were minimal, with the highly concerned group increasing from 6.049 to 6.500 and the unconcerned group improving slightly from 5.273 to 5.879. These findings indicate that low-skill workers struggled the most with adapting to technological changes, especially those who were uncertain about AI replacement.

The marginal effects analysis highlights the crucial role of skill levels in shaping adaptability. High-skill workers in the highly concerned group showed the most significant improvements in technology adoption and well-being. In contrast, low-skill workers in the unconcerned group faced declining technological engagement. These results underscore the need for targeted skill retraining programs to help low-skill workers bridge the adaptation gap and ensure more equitable responses to AI-driven technological shifts.

6. Discussion and conclusion

6.1. Discussion

This study applies a Difference-in-Differences model to examine how generative AI technology has affected Taiwan's labor market across three key periods: 2020 (pre-ChatGPT), 2022 (ChatGPT launch), and 2023 (one year post-ChatGPT). The findings reveal a worker response trajectory characterized by "short-term anxiety—mid-term adjustment—long-term divergence" in reaction to AI-driven technological changes.

Workers concerned about AI replacement made significant behavioral adjustments early in the ChatGPT adoption phase, particularly by increasing digital skill usage. This suggests that perceived substitution risk initially drove greater digital engagement. However, as learning costs rose and returns diminished, adaptation slowed, leading to a plateau in technology use for some workers. These findings highlight the importance of early adaptation efforts and show how technological

Table 7
Empirical results of marginal effects analysis under AI job displacement concerns across different skill levels.

Marginal Effects	Job Requiring Computer and Internet Use Rate								Subjective Well-Being in Tech Trends							
	Pre-Post ChatGPT Launch Year				ChatGPT Launch to One Year Later				Pre-Post ChatGPT Launch Year				ChatGPT Launch to One Year Later			
	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor	Level 1 Tech Labor	Level 2 Tech Labor	Level 3 Tech Labor	Level 4 Tech Labor
AIdisplace = 1, Post = 0	73.084	70.212	51.749	24.918	70.464	54.378	44.840	25.957	6.769	6.525	6.295	6.049	6.816	6.909	6.282	5.829
AIdisplace = 1, Post = 1	70.464	54.378	44.840	25.957	78.500	73.965	53.754	25.483	6.816	6.909	6.282	5.829	7.106	6.931	6.625	6.500
AIdisplace = 2, Post = 0	69.405	64.610	48.369	20.905	72.303	72.410	49.959	18.929	7.016	6.892	6.342	5.982	7.071	6.884	6.529	5.868
AIdisplace = 2, Post = 1	72.303	72.410	49.959	18.929	75.570	70.000	53.135	23.495	7.071	6.884	6.529	5.868	7.174	7.044	6.676	6.238
AIdisplace = 3, Post = 0	67.790	63.860	47.417	13.666	68.689	65.424	46.904	15.977	7.101	6.973	6.437	5.662	7.232	7.257	6.687	5.950
AIdisplace = 3, Post = 1	68.689	65.424	46.904	15.977	72.934	62.065	49.297	13.334	7.232	7.257	6.687	5.950	7.385	7.045	6.649	6.047
AIdisplace = 4, Post = 0	65.863	56.129	35.455	8.527	69.948	54.459	38.216	11.364	7.268	6.960	6.168	5.273	7.485	7.267	6.507	6.000
AIdisplace = 4, Post = 1	69.948	54.459	38.216	11.364	68.578	57.936	34.752	9.751	7.485	7.267	6.507	6.000	7.407	7.195	6.459	5.879

Note: The values in the table are all statistically significant at the 1 % level, indicating that technological dependence and subjective well-being aligned with technological trends for specific labor groups are highly stable and representative rather than random or coincidental outcomes.

shifts impact workers differently based on skill level.

Subjective technological well-being followed an "initial decline—mid-term recovery—long-term divergence" pattern. Early uncertainty and replacement fears caused a decline, but skill enhancement and behavioral adjustments supported recovery. While this trajectory resonates with the Technology Acceptance Model's premise that perceived usefulness and ease of use drive acceptance, it also exposes critical nuances beyond TAM's scope. The pronounced initial anxiety represents a fear-driven psychological shock that traditional acceptance models do not explicitly address, highlighting the need to incorporate early-stage stress and perceived loss of control into theory. Moreover, the observed long-term divergence in well-being suggests that adaptation pathways split based on factors such as skill level and digital readiness, indicating that one-size-fits-all models may overlook the heterogeneous nature of technological adaptation.

Notably, the empirical trajectory of "short-term anxiety—mid-term adjustment—long-term divergence" maps directly onto the three-stage dynamic adaptation model outlined earlier. In the theoretical framework, the first stage (pre-ChatGPT) is a baseline with no AI disruption, corresponding to the stable conditions observed in 2020 before ChatGPT's introduction. The second stage (initial ChatGPT integration in 2022) predicted moderate, low-cost task reallocation – reflected empirically in the immediate uptick in computer and internet use among highly concerned workers as they proactively adjusted their tasks to mitigate perceived displacement risk. The third stage (full technology diffusion by 2023) in the model involves more substantial skill realignment; this aligns with the data showing a rebound in digital tool usage and a recovery in technological well-being one year post-ChatGPT, as workers undertook deeper adaptation once the new technology became firmly integrated.

In other words, the empirical findings are evidenced by the two-step adjustment process hypothesized by the dynamic model – an early adjustment followed by a more comprehensive adaptation. The model also anticipated heterogeneity in long-run outcomes: higher-skill, digitally ready workers reach a new equilibrium of high technology use and restored well-being, whereas lower-skill or less digitally prepared workers face a plateau in adaptation. This theoretical insight helps explain the observed long-term divergence in outcomes across skill groups. By grounding the empirical results in the dynamic adaptation theory, we see a strong conceptual linkage: the workers' behavioral and psychological responses over 2020–2023 unfold in stages that mirror the model's stages of technological adaptation.

These results contribute to the literature on AI's labor market effects. Consistent with Song (2024), high-skill workers adapted quickly, leveraging new technologies for competitive advantages, while low-skill workers struggled with adaptation. Service sector workers, who rely heavily on data and intelligent tools, adjusted faster than manufacturing workers, who followed a more gradual adaptation process. These findings align with Wang et al. (2024), emphasizing the role of industrial digitization in shaping adaptive behaviors. However, because our data come from repeated cross-sections rather than a longitudinal panel, we cannot observe any worker's change in job roles or tasks over time. Thus, the differences identified across skill levels or industries reflect broad group-level patterns rather than direct evidence of individual structural changes.

Early behavioral adjustments among highly anxious workers helped mitigate substitution risks but were constrained by rising learning costs. Limited adaptability underscores the need for targeted skill retraining programs for low-skill workers. The recovery in technological well-being suggests a cyclical "learning—acceptance—adjustment" process, integrating psychological and behavioral adaptation. Thus, the initial fear-induced psychological disequilibrium is gradually resolved through adaptation efforts. Importantly, this adaptation does not follow a singular path but bifurcates into divergent trajectories. One path involves effective skill growth and successful technology integration—commonly seen among higher-skill workers or in sectors with high digital

readiness—leading to restored confidence and technological well-being. The other path, often observed among lower-skill workers or in sectors with low digital readiness, is marked by slower adjustment and continued strain, resulting in a sustained gap in technological well-being. This bifurcated adaptation pattern represents a conceptual refinement of prevailing technology acceptance models, underscoring that individual skills and contextual digital readiness can lead to fundamentally different long-term outcomes from the same AI disruption.

This study has limitations. Cross-sectional data limit the ability to track long-term changes at the individual level (for example, our design cannot directly observe changes in an individual's job typology or task composition over time), so future research could use longitudinal methods for a more comprehensive analysis. Additionally, because this research focuses exclusively on Taiwan—a high-tech economy with advanced digital infrastructure and proactive AI policies—its findings reflect a specific geographic and institutional context. Results may vary in countries with different industrial structures, technological maturity, or labor market regulations. Future research should conduct comparative cross-national studies (in other OECD economies and emerging markets) to assess the transferability of the fear–adaptation–divergence model across diverse settings.

Another limitation is that the key measure of AI concern is self-reported, which may introduce endogeneity bias. Individuals predisposed to higher anxiety or lower digital self-confidence might report greater AI-related concern and experience different adaptation outcomes, potentially confounding the analysis. While the Difference-in-Differences design helps mitigate certain biases, it cannot fully account for such unobserved heterogeneity. Therefore, any causal interpretation of the observed relationships should be made cautiously, and the findings are more suggestive than definitive given this measurement constraint. The focus on subjective technological well-being excludes broader psychological factors such as job stress and satisfaction. Future studies incorporating multidimensional indicators provide a more holistic perspective.

By adopting a dynamic perspective, this study highlights the varied impacts of AI on worker adaptation and technological well-being. The findings underscore that adaptation is gradual and requires sustained support. Addressing gaps in dynamic and heterogeneity analyses, this research provides valuable insights for policymakers and businesses to balance AI development with labor market stability, ensuring long-term worker adaptation in an era of advanced technologies.

6.2. Conclusion

This study analyzes the dynamic impact of AI technology on Taiwan's labor market, focusing on workers' behavioral adaptations and changes in subjective technological well-being. It highlights varying responses across industries and skill levels. The findings reveal an adaptive trajectory driven by AI substitution pressure, characterized by "short-term anxiety—mid-term adjustment—long-term divergence," with apparent differences across industries and skill levels.

Workers highly concerned about AI replacement showed significant increases in digital skill utilization during the early stages of AI adoption. Perceived substitution risks initially motivated this shift. However, motivation declined as learning costs rose and marginal returns diminished, leading to a plateau or cessation in technology adoption. At the same time, subjective technological well-being followed a dynamic pattern. It initially declined due to uncertainty and anxiety but gradually recovered through behavioral adjustments and psychological adaptation.

Adaptation patterns varied across industries and skill levels. Service sector workers, who rely heavily on digital tools, adapted more quickly and showed greater flexibility in early technology adoption. In contrast, manufacturing workers exhibited more stable and gradual technological changes. High-skill workers swiftly leveraged technological demands, enhancing digital skill utilization and improving subjective technological well-being. Meanwhile, low-skill workers struggled with adaptation, underscoring the need for targeted skill retraining and support measures.

This study contributes to the literature in several ways. First, it adopts a dynamic perspective, uncovering the phased impacts of AI on worker behavior and psychological adaptation, addressing gaps in research on technological disruptions. Second, subsample analysis provides deeper insights into heterogeneous responses across industries and skill levels, revealing distinct adaptation pathways. Third, it examines the interaction between behavioral adjustments and psychological adaptation, offering theoretical and empirical insights into AI's labor market effects.

The findings provide important implications for policymakers and businesses. Governments should implement skill retraining programs, particularly for low-skill workers, to close the digital divide and improve technological adaptability. Enterprises should establish human-machine collaboration models, enhance digital skill training, and provide psychological support to ease substitution pressures and facilitate AI integration. At the industry level, efforts should focus on strengthening the digital infrastructure in the service sector to support rapid adaptation while promoting steady technological upgrades in manufacturing to ensure balanced development and labor market stability.

In conclusion, this study highlights the challenges and opportunities of generative AI adoption. It underscores the adaptive advantage of high-skill workers and the urgent need for skill training among low-skill workers. However, these conclusions are drawn from Taiwan's unique context and may not be fully generalized to other settings. Economies with different digital landscapes or institutional frameworks could experience the fear–adaptation–divergence trajectory differently, underscoring the need for comparative cross-national research to validate and extend this model. Addressing these transformations requires collaboration between governments and businesses to promote skill training, psychological support, and digital inclusion policies. These initiatives will help workers adapt more smoothly to technological transitions, fostering labor market stability and sustainable economic growth.

Data availability statement

The data that support the findings of this study are openly available in the Survey Research Data Archive of Academia Sinica, conducted by the National Development Council and the Ministry of Digital Affairs of the Taiwan government, at the following links:
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Ethical approval is not applicable to this manuscript.

Declaration of the use of AI

The author declares that no AI assisted technologies were used during any stage in the preparation of this article.

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